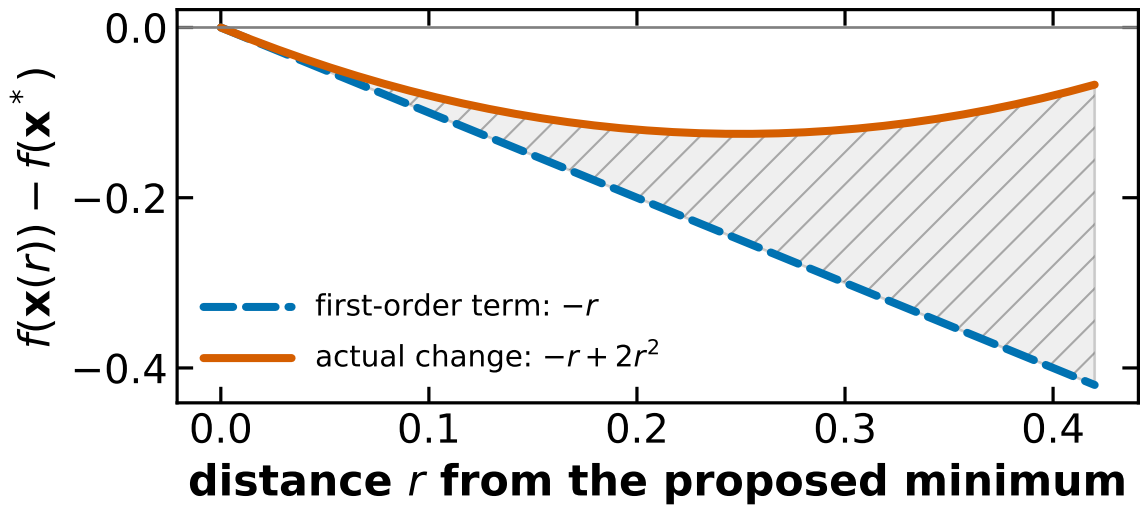
0.0
−0.2
−0.4
$f(\mathbf{x}(r)) - f(\mathbf{x}^*)$
first-order term: $-r$
actual change: $-r + 2r^2$
0.0
0.1
0.2
0.3
0.4
distance r from the proposed minimum